# Tamanna Bhrigunath

+91-8233535365 | bhrigunathtamanna@gmail.com | linkedin.com/in/tamanna-bhrigunath | github.com/bhrigu136 |
leetcode.com/u/XuMspNFAS8 | portfolio-fawn-xi-83.vercel.app

## Summary

B.Tech graduate in AI & Data Science with hands-on experience across the full ML lifecycle - data cleaning, model building, evaluation, and deployment. Currently building production ERP modules (attendance, face verification, payroll) at Nessco that serve real users daily. Interned at Celebal Technologies where I shipped two end-to-end ML systems.

## Experience

**Software Engineering Intern** 2026 – Present
*Nessco Paper Cup Machine* *Jaipur, India*

- Built and shipped three production ERP modules — attendance tracking, biometric face check-in, and payroll — used by the organization's employees on a daily basis.
- Designed a race-safe attendance punch system using idempotency keys and a `[userId, date]` unique constraint, making duplicate records impossible even under simultaneous retries.
- Implemented browser-side face verification using **face-api.js**: captures 3 frames for passive liveness detection, generates a 128-float facial descriptor, and matches it server-side via Euclidean distance — raw biometric data never leaves the device.
- Built a live payroll engine (Next.js + Prisma + PostgreSQL) that recomputes payslips on every read, covering PF, ESI, PT, TDS, pro-rated salaries, and overtime multipliers — no nightly batch jobs.

**Data Science Intern** June 2025 – August 2025
*Celebal Technologies* *Remote*

- Built two end-to-end ML pipelines from raw data to deployed Streamlit apps: a house price prediction system and a credit risk classifier.
- Developed a credit risk classification model using **Random Forest** on the German Credit dataset; resolved class imbalance via resampling and achieved **˜80% accuracy**, evaluated with precision, recall, and F1-score instead of accuracy alone.
- Benchmarked Linear Regression, Random Forest, and SVM for house price prediction using cross-validation, selecting the model with the best generalization on unseen data.
- Performed feature engineering and EDA using pandas, NumPy, Matplotlib, and Seaborn; serialized trained models with Joblib for production inference.

## Projects

**HiveFlow: Collaborative Task Manager** | *Flask · SQLAlchemy · OAuth 2.0 · PostgreSQL · Render* | GitHub | Live 2025

- Designed and deployed a full-stack multi-tenant task management platform supporting organizations, Kanban boards, role-based access, CSV export, and email OTP-based password recovery.
- Integrated Google Calendar API (OAuth 2.0) so task creation, updates, and deletions automatically sync to users' calendars; handled token refresh and external API failures gracefully.
- Modeled relational data with SQLAlchemy enforcing strict user-level task isolation; deployed on Render with Gunicorn and PostgreSQL in production.

**Movie Recommendation System** | *scikit-learn · CountVectorizer · TMDB API · Streamlit* | GitHub | Live 2025

- Built a content-based recommender system that merges genres, cast, director, and plot into a single feature vector and ranks top-5 similar titles using cosine similarity.
- Fetched real-time movie posters via TMDB API and delivered results through an interactive Streamlit interface.

**Credit Scoring System** | *Random Forest · scikit-learn · Joblib · Streamlit* | GitHub | Live 2025

- Classified loan applicants as good/bad credit risk (**˜80% accuracy**) using Random Forest on the German Credit dataset; resolved class imbalance with resampling and analyzed feature importance for decision explainability.

**House Price Prediction** | *Regression · SVM · Cross-Validation · scikit-learn · Streamlit* | GitHub | Live 2025

- Compared Linear Regression, Random Forest, and SVM side-by-side using k-fold cross-validation to identify the best-generalizing model, then deployed it as a real-time Streamlit app.

## Technical Skills

**Languages**: Python, Java, SQL
**Machine Learning**: Regression, Classification, Random Forest, SVM, Feature Engineering, Model Evaluation, Imbalanced Data Handling
**Generative AI**(learning): LLM APIs (Gemini), Prompt Engineering, Context Injection, RAG
**ML Libraries**: scikit-learn, pandas, NumPy, Matplotlib, Seaborn, Joblib, Pickle
**Backend & Web**: Flask, REST APIs, SQLAlchemy, Flask-Login, React, TypeScript, HTML, Tailwind CSS, Next.js
**Databases**: PostgreSQL, MySQL
**APIs & Integrations:** Google Calendar API, OAuth 2.0, TMDB API
**Deployment & Tools**: Streamlit, Render, Gunicorn, Vercel, Git, Jupyter Notebook, VS Code

## Education

**Poornima College of Engineering** — Jaipur, India
*B.Tech in Artificial Intelligence & Data Science* — *2022 – 2026*

- Relevant Coursework: Machine Learning, Data Mining, Image Processing, Information Retrieval, DSA, Computer Networks, DBMS, Operating Systems, OOP

## Achievements

**Runner-up — AADHAR-XI Technical Project Competition (2023)**: Awarded second place at a college-level Technical Project Competition and Exhibition organised by Zircon Club, Poornima College of Engineering.